Beyond the Dailey-Townes model: chemical information from the Electric Field Gradient in a Relativistic framework

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HAMP-vQED

Highly Accurate Molecular Properties using variational Quantum Electrodynamics







HAMP

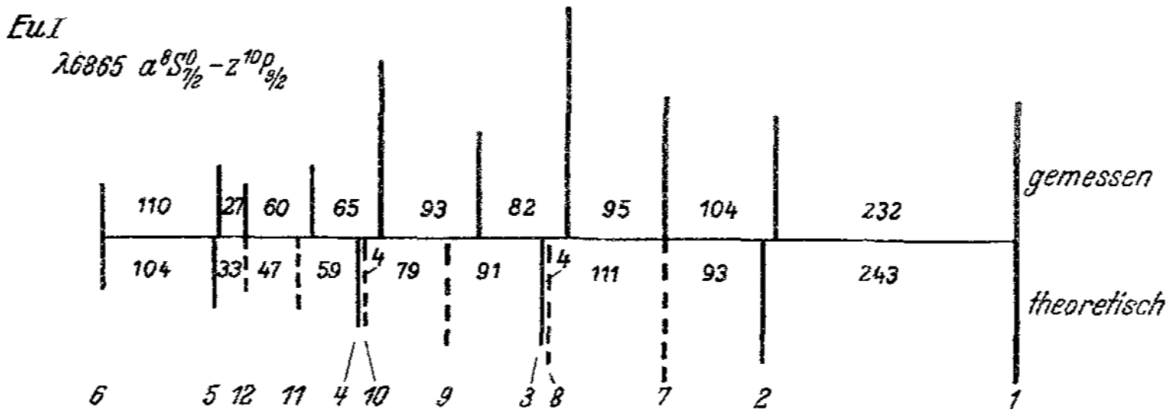
- Computational protocol for the **highly accurate computation of molecular properties**.
- **High-level relativistic coupled cluster theory**.
  - **CCSD(T): golden standard in quantum chemistry**

vQED

- Heavy elements need the **special theory of relativity** to be understood.
- HAMP-vQED project will investigate the possible role of **quantum electrodynamics (QED)** in chemistry.

Introduction

- **Schüler and Schmidt (1935)**[1] observed deviations from the Landé interval rule associated with the nuclear magnetic dipole moment in the hyperfine splittings of the atomic spectra of ^{151}Eu and ^{153}Eu.

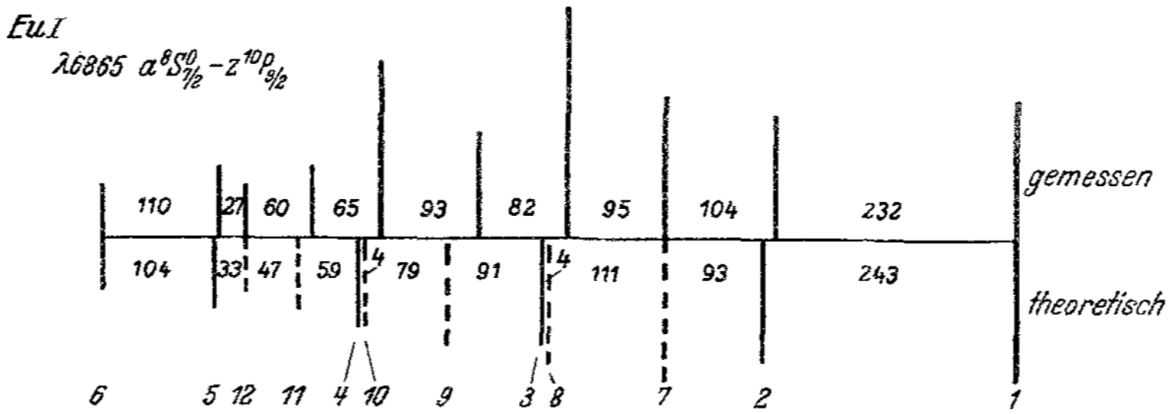


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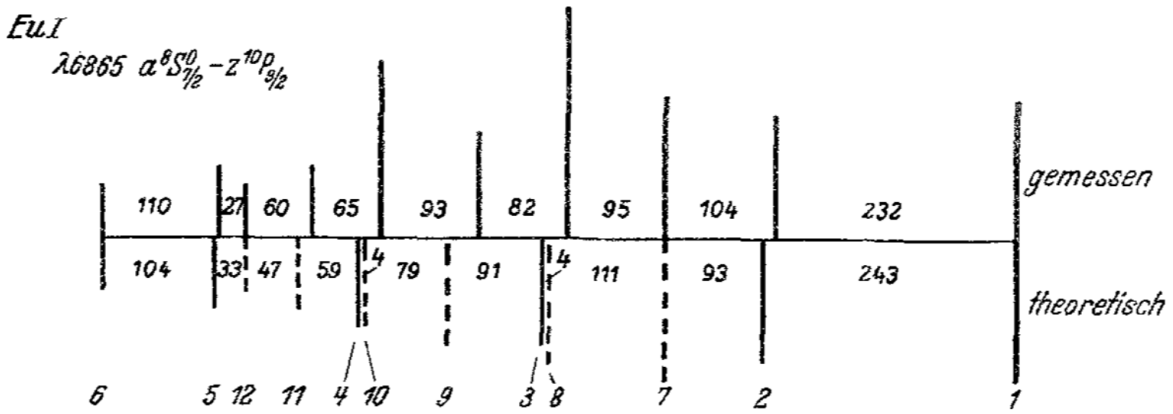
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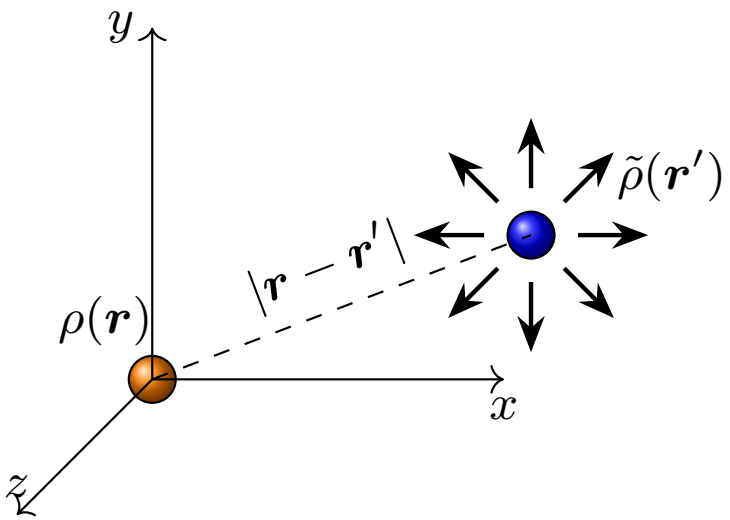
- Weeks later, **Casimir**[2] formulates the theory necessary to understand the phenomenon.
  - Nuclei with $I > 1/2$ possess a **nuclear electric quadrupole moment** which couples with the **electric field gradient** generated by the other charges.

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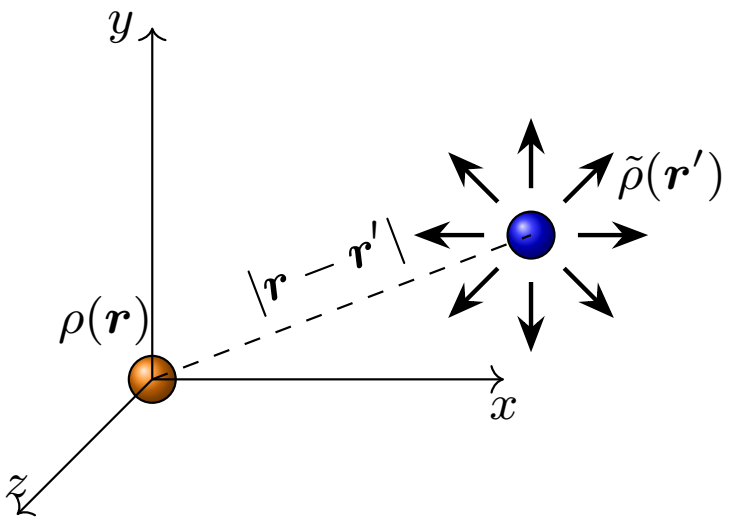
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Multipole expansion



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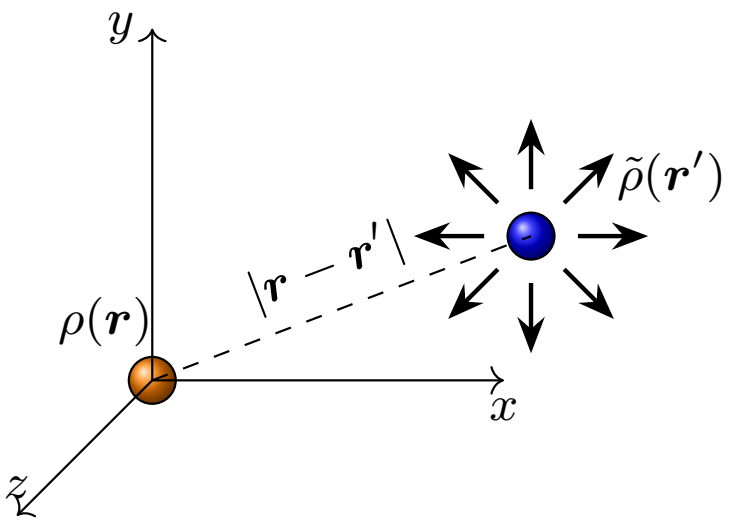


- The potential that we feel at point $\boldsymbol{r}$:

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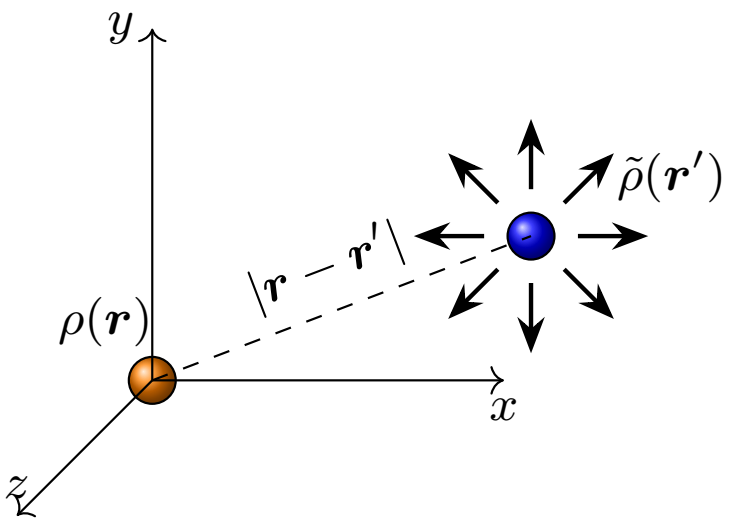
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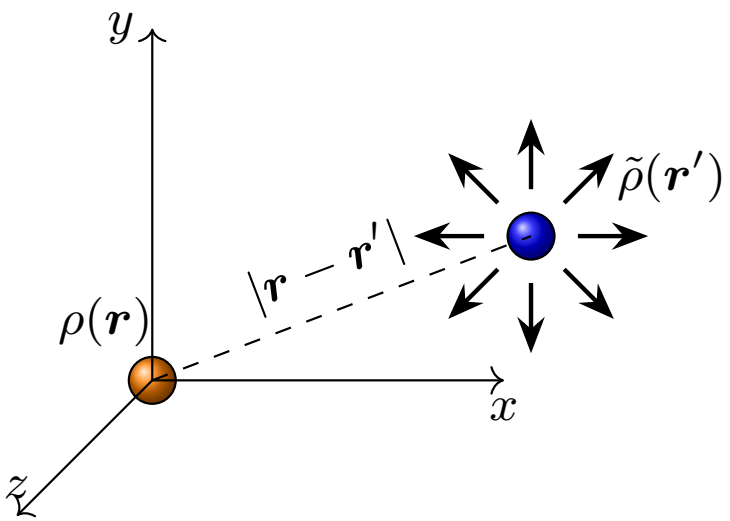
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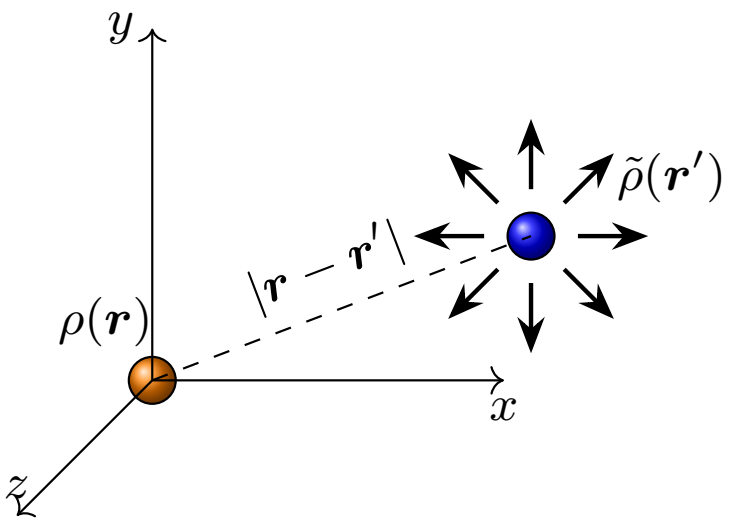
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- **Quadrupole interaction energy:**

$$\boxed{E_{E_2} = -\frac{1}{2} \sum_{\alpha\beta} Q_{\alpha\beta}(\boldsymbol{r}_0) \mathcal{E}^{[1]}_{\alpha;\beta}(\boldsymbol{r}_0)}$$

Origin of the quadrupole interaction

Traceless quadrupole moments

- We have defined the electric quadrupole moment tensor $\boldsymbol{Q}_0$ and the electric field gradient tensor $\boldsymbol{\mathcal{E}}_0^{[1]}$.

$$Q_{\alpha\beta}(\boldsymbol{r}_0) = \int \rho(\boldsymbol{r})(r_\alpha - r_{0,\alpha})(r_\beta - r_{0,\beta})d\boldsymbol{r} \qquad \mathcal{E}^{[1]}_{\alpha;\beta}(\boldsymbol{r}_0) = \left.\frac{\partial \mathcal{E}_\alpha}{\partial r_\beta}\right|_{\boldsymbol{r}=\boldsymbol{r}_0} \tag{5}$$

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- We can switch to the **traceless quadrupole moments**:

$$\Theta_{\alpha\beta}(\boldsymbol{r}_0) = \frac{3}{2}\left(Q_{\alpha\beta}(\boldsymbol{r}_0) - \frac{1}{3}\delta_{\alpha\beta}\sum_\gamma Q_{\gamma\gamma}(\boldsymbol{r}_0)\right) \tag{6}$$

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- The interaction energy in terms of traceless quadrupole moments becomes:

$$E_{E_2} = -\frac{1}{3}\sum_{\alpha\beta}\Theta_{\alpha\beta}(\boldsymbol{r}_0)\mathcal{E}^{[1]}_{\alpha;\beta}(\boldsymbol{r}_0) - \frac{1}{6}\sum_{\gamma,\alpha} Q_{\gamma\gamma}(\boldsymbol{r}_0)\mathcal{E}^{[1]}_{\alpha;\alpha}(\boldsymbol{r}_0) \tag{7}$$

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  - **Poisson term**: from Gauss law is strictly zero if the charge distribution generating the electric field is outside the expansion point:

$$(\boldsymbol{\nabla}\cdot\boldsymbol{\mathcal{E}})_{\boldsymbol{r}=\boldsymbol{r}_0} = \sum_\alpha \mathcal{E}^{[1]}_{\alpha;\alpha} = \frac{\rho(\boldsymbol{r}_0)}{\varepsilon_0} = 0$$

Origin of the quadrupole interaction

Nuclear quadrupole moment

- If we consider a charge distribution generated by a **nucleus**, the traceless nuclear electric quadrupole moment is given by:

$$\Theta_{\alpha\beta} = \int \rho_n(\boldsymbol{r})\hat{\Theta}_{\alpha\beta} d\boldsymbol{r} \ , \ \hat{\Theta}_{\alpha\beta} = \frac{1}{2}(3\hat{r}_\alpha \hat{r}_\beta - \delta_{\alpha\beta}\hat{r}^2) \tag{8}$$

where for convenience we have located the nucleus at the origin.

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$$\Theta_{\alpha\beta} = C\,\langle I, m_I|\frac{1}{2}(3\hat{I}_\alpha\hat{I}_\beta - \delta_{\alpha\beta}\hat{I}^2)|I, m_I\rangle \ , \ C = \frac{eQ}{\hbar^2 I(2I-1)} \tag{9}$$

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  - In our case **the Poisson term is not zero** since the electronic charge density $\rho(\boldsymbol{r}_0)$ can not be expected to be zero at the nuclear origin.

Origin of the quadrupole interaction

Nuclear quadrupole coupling

- Using angular momentum theory, one can show that in atoms and linear systems the interaction energy is expressed as:

$$E_{E_2}(I, m_I) = \frac{e^2 qQ}{4I(2I-1)}[3m_I^2 - I(I+1)] - \frac{1}{6\varepsilon_0}\frac{eQ(I+1)}{(2I-1)}\rho(\boldsymbol{r}_0) \tag{11}$$

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$$\text{NQCC} = \frac{e^2 qQ}{h} \ , \ \text{NQCC[MHz]} = 234.9647 \times Q[b] \times q[E_h/ea_0^2] \tag{14}$$

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  - **What about chemical information?**

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Chemical information from the electric field gradient

- The electric field gradient at nucleus K is due to a **nuclear** and **electronic** contribution:

$$\langle e\hat{q}_K \rangle = \underbrace{-\frac{e}{4\pi\epsilon_0} \sum_{i=1}^{\mathrm{N_{occ}}} \left\langle \frac{3\hat{z}_{i;K}^2 - \hat{r}_{i;K}^2}{\hat{r}_{i;K}^5} \right\rangle}_{\textbf{electronic}} + \underbrace{\sum_{A \neq K} \frac{Z_A e}{4\pi\epsilon_0} \left[\frac{3Z_{A;K}^2 - R_{A;K}^2}{R_{A;K}^5} \right]}_{\textbf{nuclear}} \tag{15}$$

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- The electronic contribution:
  - Samples the inner tails of valence orbitals;
  - Sensitive to small deformation close to the nucleus;
  - Sensitive to electronic correlation and relativistic effects;

Projection analysis

S. Dubillard, J.-B. Rota, T. Saue and K.Fægri, J. Chem. Phys. 124 (2006) 154307

- MOs are expanded in a set of **occupied** atomic orbitals $|\psi_p^A\rangle$ of the **constituent atoms/fragments** calculated at SCF level:

$$|\Psi_i^{\mathrm{MO}}\rangle = \sum_A \sum_{p \in A} c_{p,i}^A \, |\psi_p^A\rangle + |\psi_i^{\mathrm{pol}}\rangle \tag{16}$$

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- The electronic contribution now reads

$$\langle e\hat{q}_K^{\mathrm{el}}\rangle = \sum_i^{\mathrm{Nocc}} \langle \Psi_i^{\mathrm{MO}} | e\hat{q}_K^{\mathrm{el}} | \Psi_i^{\mathrm{MO}}\rangle = \sum_A \langle e\hat{q}_K^{\mathrm{el}}\rangle_{\mathrm{intra}}^{(A)} + \sum_A \sum_{B \neq A} \langle e\hat{q}_K^{\mathrm{el}}\rangle_{\mathrm{inter}}^{(A,B)} + \langle e\hat{q}_K^{\mathrm{el}}\rangle_{\mathrm{pol}} \tag{17}$$

Projection analysis

The intra-atomic contribution at center K

$$\langle e\hat{q}_K^{\mathrm{el}} \rangle_{\mathrm{intra}}^{(K)} = \underbrace{\sum_{p \in K} D_{pp}^{KK} \, \langle \psi_p^K | e\hat{q}_K^{\mathrm{el}} | \psi_p^K \rangle}_{\textbf{principal}} + \underbrace{\sum_{p \in K} \sum_{(q \neq p) \in K} D_{pq}^{KK} \, \langle \psi_p^K | e\hat{q}_K^{\mathrm{el}} | \psi_q^K \rangle}_{\textbf{hybridization}} \tag{18}$$

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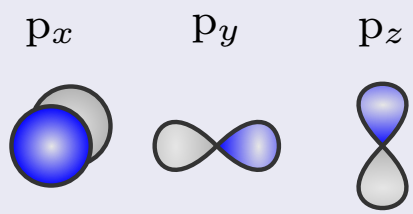
  - **s orbitals do not contribute to the electric field gradient.**
  - **What about the other orbitals?**

Projection analysis

The intra-atomic contribution at center K

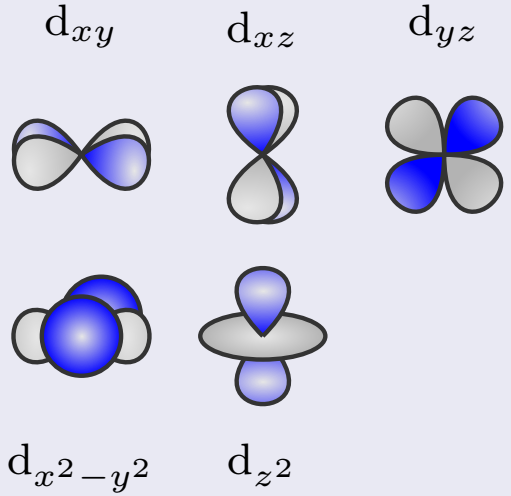
p orbitals

$$\langle p_m | e\hat{q}_K^{\mathrm{el}} | p_m \rangle = -\frac{e}{4\pi\varepsilon_0} \langle r^{-3} \rangle \times \begin{cases} \frac{4}{5} & ;\ m = 0 \\ -\frac{2}{5} & ;\ |m| = 1 \end{cases}$$



d orbitals

$$\langle d_m | e\hat{q}_K^{\mathrm{el}} | d_m \rangle = -\frac{e}{4\pi\varepsilon_0} \langle r^{-3} \rangle \times \begin{cases} -\frac{2}{7} & ;\ m = 0 \\ -\frac{1}{7} & ;\ |m| = 1 \\ \frac{2}{7} & ;|m| = 2 \end{cases}$$



Projection analysis

Albrecht Unsöld, "Beiträge zur Quantenmechanik der Atome". In: Annalen der Physik 387 (3) (1949), pp. 355–393.

Unsöld theorem

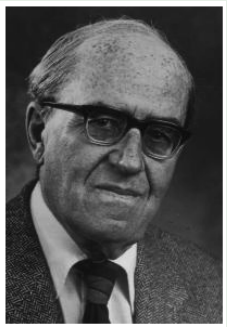


Figure 1: Albrecht Unsöld (1905-1995)

"*Any complete atomic shell is spherically symmetric*"

$$\sum_{m=-\ell}^{\ell} |Y_{\ell m}(\theta,\phi)|^2 = \frac{2\ell+1}{4\pi} \quad \Rightarrow \quad \sum_{m=-\ell}^{\ell} \langle Y_{\ell m}|Y_{20}|Y_{\ell m}\rangle_{\theta,\phi} = 0.$$

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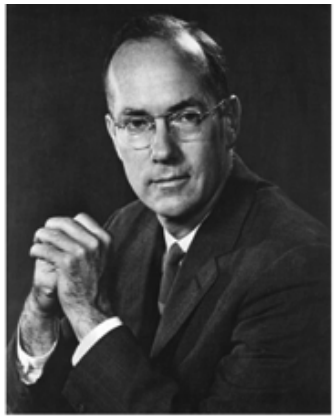
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1. Core orbitals do not contribute to the electric field gradient.
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3. Origin of the Dailey-Townes model.

Dailey-Townes model

C.H. Townes and B. P. Dailey, J. Chem. Phys. 17 (1949) 782





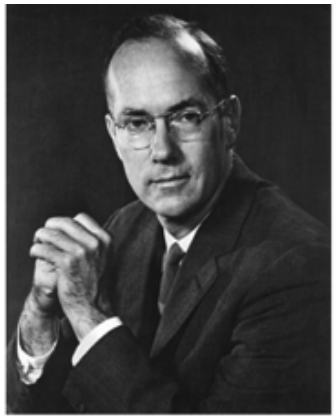
The magnitude of q at a nucleus depends almost entirely on the way in which the lowest available p-type orbits are occupied by valence electrons surrounding the nucleus. An s-type orbit or any closed shell of electrons is spherically symmetric and hence gives no contribution to the variation of energy with nuclear orientation or to $\partial^2 V/\partial z^2$ at the nucleus.

Figure 2: Benjamin Peter Dailey (1919 - 2021) Charles Hard Townes (1915 – 2015)

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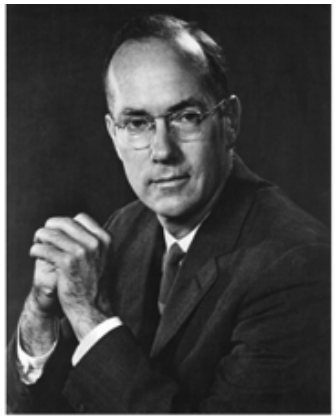
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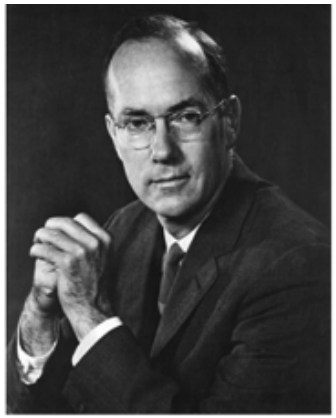
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Dailey-Townes model

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- Dailey-Townes model in the language of projection analysis:

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  - What about **relativistic effects?**

Computational details

www.diracprogram.org

P rogram
• for
A tomic
• and
M olecular

D irect
I terative
R elativistic
A ll-electron
C alculations



- 4-component Kohn-Sham calculations (PBE/dyall.3zp and dyall.v3z for U) based on the Dirac-Coulomb Hamiltonian.
- Unless otherwise stated, spin-orbit interaction was eliminated.

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

- We recall:

$$\begin{aligned} \langle e\hat{q}_{\mathrm{Cl}} \rangle &= \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle + e\hat{q}^{\mathrm{nucl}}_{\mathrm{Cl}} \\ &= \sum_{A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{intra}} + \sum_{A,B\neq A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A,B)}_{\mathrm{inter}} + \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle_{\mathrm{pol}} + e\hat{q}^{\mathrm{nucl}}_{\mathrm{Cl}} \end{aligned} \tag{24}$$

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		EFG
intra-atomic		

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			EFG
intra-atomic	Cl	Principal	5.890
		Hybridization	-0.570

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

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			EFG
intra-atomic	Cl	Principal	5.890
		Hybridization	-0.570
	I	Principal	-1.212
		Hybridization	0.00993

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

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$$\begin{aligned} \langle e\hat{q}_{\mathrm{Cl}} \rangle &= \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle + e\hat{q}^{\mathrm{nucl}}_{\mathrm{Cl}} \\ &= \sum_{A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{intra}} + \sum_{A, B \neq A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A,B)}_{\mathrm{inter}} + \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle_{\mathrm{pol}} + e\hat{q}^{\mathrm{nucl}}_{\mathrm{Cl}} \end{aligned} \tag{24}$$

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	I	Principal	-1.212
		Hybridization	0.00993
inter-atomic	Cl-I		-0.0166

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

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		Hybridization	-0.570
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		Hybridization	0.00993
inter-atomic	Cl-I		-0.0166
Polarization			-0.619

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

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			EFG
intra-atomic	Cl	Principal	5.890
		Hybridization	-0.570
	I	Principal	-1.212
		Hybridization	0.00993
inter-atomic	Cl-I		-0.0166
Polarization			-0.619
Total electronic			3.482

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

- We recall:

$$\begin{aligned} \langle e\hat{q}_{\mathrm{Cl}} \rangle &= \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle + e\hat{q}^{\mathrm{nucl}}_{\mathrm{Cl}} \\ &= \sum_{A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{intra}} + \sum_{A, B \neq A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A,B)}_{\mathrm{inter}} + \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle_{\mathrm{pol}} + e\hat{q}^{\mathrm{nucl}}_{\mathrm{Cl}} \end{aligned} \tag{24}$$

			EFG
intra-atomic	Cl	Principal	5.890
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	I	Principal	-1.212
		Hybridization	0.00993
inter-atomic	Cl-I		-0.0166
Polarization			-0.619
Total electronic			3.482
Total nuclear			1.174

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

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			EFG
intra-atomic	Cl	Principal	5.890
		Hybridization	-0.570
	I	Principal	-1.212
		Hybridization	0.00993
inter-atomic	Cl-I		-0.0166
Polarization			-0.619
Total electronic			3.482
Total nuclear			1.174
Total EFG			4.737

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $\mathrm{I}^{35}\mathrm{Cl}$

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$$\begin{aligned}\langle e\hat{q}_{\mathrm{Cl}} \rangle &= \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle + e\hat{q}^{\mathrm{nucl}}_{\mathrm{Cl}} \\ &= \sum_{A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{intra}} + \sum_{A,B\neq A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A,B)}_{\mathrm{inter}} + \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle_{\mathrm{pol}} + e\hat{q}^{\mathrm{nucl}}_{\mathrm{Cl}}\end{aligned} \tag{24}$$

			EFG
intra-atomic	Cl	Principal	**5.890**
		Hybridization	-0.570
	I	Principal	-1.212
		Hybridization	0.00993
inter-atomic	Cl-I		-0.0166
Polarization			-0.619
Total electronic			3.482
Total nuclear			1.174
Total EFG			4.737

DT model assumes to consider only the contributions of the AOs of the atom investigated

It works!

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $I^{35}Cl$

- We recall:

$$\begin{aligned} \langle e\hat{q}_{\mathrm{Cl}} \rangle &= \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle + e\hat{q}^{\mathrm{nucl}}_{\mathrm{Cl}} \\ &= \sum_{A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A)}_{\mathrm{intra}} + \sum_{A, B \neq A} \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(A,B)}_{\mathrm{inter}} + \langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle_{\mathrm{pol}} + e\hat{q}^{\mathrm{nucl}}_{\mathrm{Cl}} \end{aligned} \tag{24}$$

			EFG
intra-atomic	Cl	Principal	5.890
		Hybridization	-0.570
	I	Principal	**-1.212**
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inter-atomic	Cl-I		-0.0166
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			EFG
intra-atomic	Cl	Principal	5.890
		Hybridization	-0.570
	I	Principal	**-1.212**
		Hybridization	0.00993
inter-atomic	Cl-I		-0.0166
Polarization			-0.619
Total electronic			3.482
Total nuclear			**1.174**
Total EFG			4.737

DT model assumes to neglect the contributions of the AOs of the other atoms

$$\langle e\hat{q}_{\mathrm{Cl}} \rangle^{(\mathrm{I})}_{\mathrm{princ}} + e\hat{q}^{\mathrm{I}}_{\mathrm{Cl}} \simeq 0$$

It works!

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $I^{35}Cl$

Dailey-Townes model for $I^{35}Cl$

- Origin of $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl})}_{\mathrm{princ}}$

Expectation value	EFG	Σ
$\langle 1s\|\hat{q}^{\mathrm{el}}_{\mathrm{Cl}}\|1s\rangle$	0	0
$\langle 2s\|\hat{q}^{\mathrm{el}}_{\mathrm{Cl}}\|2s\rangle$	0	0
$\langle 2p_x\|\hat{q}^{\mathrm{el}}_{\mathrm{Cl}}\|2p_x\rangle$	81.586	
$\langle 2p_y\|\hat{q}^{\mathrm{el}}_{\mathrm{Cl}}\|2p_y\rangle$	81.586	
$\langle 2p_z\|\hat{q}^{\mathrm{el}}_{\mathrm{Cl}}\|2p_z\rangle$	-163.186	
		$\simeq 0$
$\langle 3s\|\hat{q}^{\mathrm{el}}_{\mathrm{Cl}}\|3s\rangle$	0	0
$\langle 3p_x\|\hat{q}^{\mathrm{el}}_{\mathrm{Cl}}\|3p_x\rangle$	6.101	
$\langle 3p_y\|\hat{q}^{\mathrm{el}}_{\mathrm{Cl}}\|3p_y\rangle$	6.101	
$\langle 3p_z\|\hat{q}^{\mathrm{el}}_{\mathrm{Cl}}\|3p_z\rangle$	-6.297	
		5.905

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $I^{35}Cl$

Dailey-Townes model for $I^{35}Cl$

- Origin of $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl})}_{\mathrm{princ}}$

Expectation value	EFG	Σ
$\langle 1s\|\hat{q}^{\mathrm{el}}_{\mathrm{Cl}}\|1s\rangle$	0	0
$\langle 2s\|\hat{q}^{\mathrm{el}}_{\mathrm{Cl}}\|2s\rangle$	0	0
$\langle 2p_x\|\hat{q}^{\mathrm{el}}_{\mathrm{Cl}}\|2p_x\rangle$	81.586	
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		5.905

1. As predicted by Dailey and Townes, s orbitals do not contribute to the electric field gradient.

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $I^{35}Cl$

Dailey-Townes model for $I^{35}Cl$

- Origin of $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl})}_{\mathrm{princ}}$

Expectation value	EFG	Σ
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		5.905

1. As predicted by Dailey and Townes, s orbitals do not contribute to the electric field gradient.
2. In agreement with Unsöld theorem, shells fully occupied do not contribute to the electric field gradient.

Analysis of $\langle e\hat{q}_{\mathrm{Cl}} \rangle$ in $I^{35}Cl$

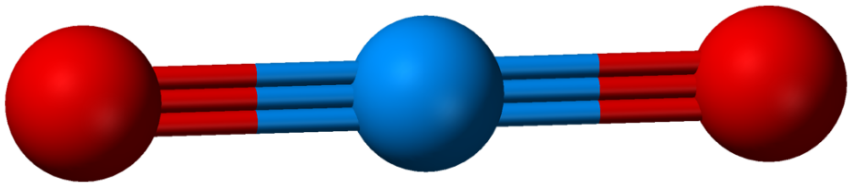
Dailey-Townes model for $I^{35}Cl$

- Origin of $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{Cl}} \rangle^{(\mathrm{Cl})}_{\mathrm{princ}}$

Expectation value	EFG	Σ
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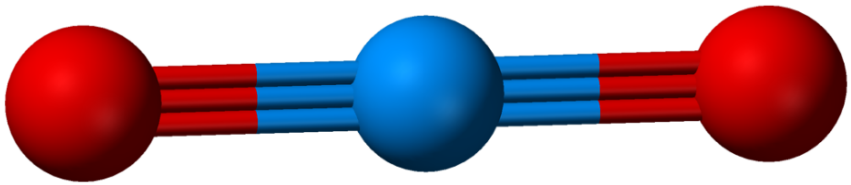
1. As predicted by Dailey and Townes, s orbitals do not contribute to the electric field gradient.
2. In agreement with Unsöld theorem, shells fully occupied do not contribute to the electric field gradient.
3. Only the valence p orbitals contribute to the electric field gradient.

Beyond the Dailey-Townes model: EFG of ^{235}U ($I = 7/2$) in uranyl $[\mathrm{O} \equiv \mathrm{U} \equiv \mathrm{O}]^{2+}$



- Uranium atom in his ground state: $[\mathrm{Hg}]6p^6 5f^3 6\mathrm{d}^1 7s^2$

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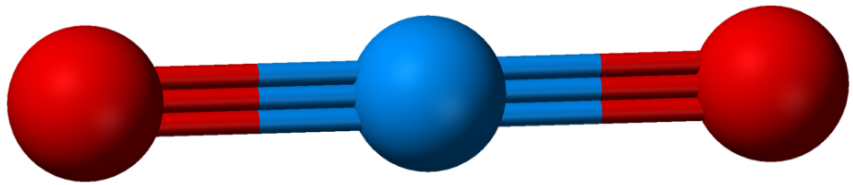


- Uranium atom in his ground state: $[\mathrm{Hg}]6p^6 5f^3 6\mathrm{d}^1 7s^2$

Ionic picture

- $\mathrm{Q_U}$=+6
- Electronic configuration: $[\mathrm{Hg}]6p^6$.

Beyond the Dailey-Townes model: EFG of ^{235}U ($I = 7/2$) in uranyl $[\mathrm{O} \equiv \mathrm{U} \equiv \mathrm{O}]^{2+}$

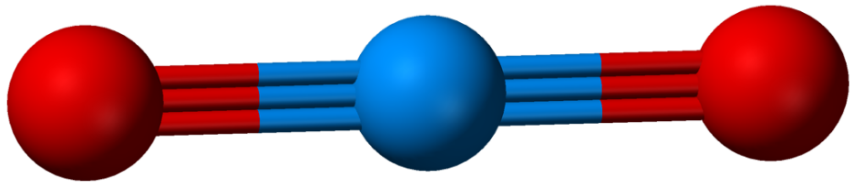


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Projection analysis picture

- Q_U=+2.46
- Electronic configuration: $[\mathrm{Hg}]6p^{5.72}\ 5\mathrm{f}^{2.74} 6d^{1.05} 7\mathrm{s}^{0.05}$.

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- Uranium atom in his ground state: $[\mathrm{Hg}]6p^6 5f^3 6\mathrm{d}^1 7s^2$

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  ! $6p$ hole

Uranyl: gross populations from projection analysis in canonical orbitals

	Energy (E_h)	6s	6p	5f	6d	7s	O_1 2s	O_1 2p	O_2 2s	O_2 2p
σ_1	-2.343	1.93								
σ_2	-1.644		1.08	0.035			0.35	0.09	0.35	0.09
π_1	-1.432		1.96							
π_2	-1.432		1.96							
σ_3	-1.374	0.04			0.15		0.85		0.85	
σ_4	-1.124		0.53	0.08			0.59	0.10	0.59	0.10
π_3	-0.863				0.39			0.77		0.77
π_4	-0.863				0.39			0.77		0.77
π_5	-0.851			0.73				0.62		0.62
π_6	-0.851			0.69				0.62		0.62
σ_5	-0.833	0.02			0.07	0.04	0.08	0.82	0.08	0.82
σ_6	-0.805		0.14	1.16			0.02	0.32	0.02	0.32
Sum		1.99	5.68	2.70	1.00	0.05	1.89	4.10	1.89	4.10

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	Energy (E_h)	6s	6p	5f	6d	7s	O_1 2s	O_1 2p	O_2 2s	O_2 2p
σ_1	-2.343	1.93								
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π_1	-1.432		1.96							
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σ_3	-1.374	0.04			0.15		0.85		0.85	
σ_4	-1.124		0.53	0.08			0.59	0.10	0.59	0.10
π_3	-0.863				0.39			0.77		0.77
π_4	-0.863				0.39			0.77		0.77
π_5	-0.851			0.73				0.62		0.62
π_6	-0.851			0.69				0.62		0.62
σ_5	-0.833	0.02			0.07	0.04	0.08	0.82	0.08	0.82
σ_6	-0.805		0.14	1.16			0.02	0.32	0.02	0.32
Sum		1.99	5.68	2.70	1.00	0.05	1.89	4.10	1.89	4.10

- Canonical MOs picture shows us a strong participation of U($6p$) in the chemical bond.

Uranyl: gross populations from projection analysis in localized orbitals

	Energy (E_h)	6s	6p	5f	6d	7s	O_1 2s	O_1 2p	O_2 2s	O_2 2p
σ_1	-3.429	1.02	0.79	0.16						
σ_2	-3.429	1.02	0.79	0.16						
π_1	-2.827		1.99							
π_2	-2.827		1.99							
σ_3	-2.577						1.92			
σ_4	-2.577								1.92	
σ_5	-2.116		0.11	0.39				1.32		
σ_6	-2.116		0.11	0.39						1.32
π_3	-1.733			0.34	0.19			1.40		
π_4	-1.733			0.34	0.19			1.40		
π_5	-1.733			0.34	0.19					1.40
π_6	-1.733			0.34	0.19					1.40
Sum		2.04	5.78	2.48	0.78	0.00	1.92	4.13	1.92	4.13

Uranyl: gross populations from projection analysis in localized orbitals

	Energy (E_h)	6s	6p	5f	6d	7s	O_1 2s	O_1 2p	O_2 2s	O_2 2p
σ_1	-3.429	1.02	0.79	0.16						
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π_1	-2.827		1.99							
π_2	-2.827		1.99							
σ_3	-2.577						1.92			
σ_4	-2.577								1.92	
σ_5	-2.116		0.11	0.39				1.32		
σ_6	-2.116		0.11	0.39						1.32
π_3	-1.733			0.34	0.19			1.40		
π_4	-1.733			0.34	0.19			1.40		
π_5	-1.733			0.34	0.19					1.40
π_6	-1.733			0.34	0.19					1.40
Sum		2.04	5.78	2.48	0.78	0.00	1.92	4.13	1.92	4.13

- Localization leads to a **decrease** in U($6p$)-O($2p$) interaction.

Interruption: relativistic effects

Scalar relativistic effects

- The relativistic mass:

$$m_{\mathrm{rel}} = \gamma m = \frac{m}{\sqrt{1 - \frac{v^2}{c^2}}}$$

- The Bohr radius:

$$a_0^{\mathrm{rel}} = \frac{4\pi\varepsilon_0\hbar^2}{m_{\mathrm{rel}}}$$

- s and p orbital are **contracted** and d and f orbitals are **expanded** indirectly.

Interruption: relativistic effects

Scalar relativistic effects

- The relativistic mass:

$$m_{\mathrm{rel}} = \gamma m = \frac{m}{\sqrt{1 - \frac{v^2}{c^2}}}$$

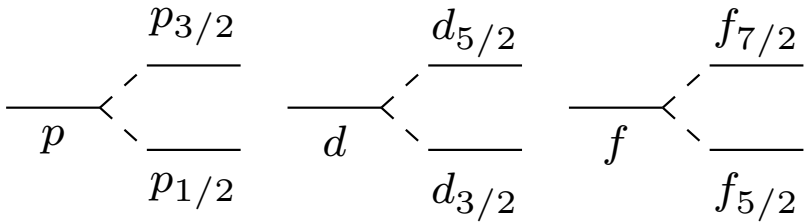
- The Bohr radius:

$$a_0^{\mathrm{rel}} = \frac{4\pi\varepsilon_0\hbar^2}{m_{\mathrm{rel}}}$$

- s and p orbital are **contracted** and d and f orbitals are **expanded** indirectly.

Spin-orbit interaction

- Interaction of the electron spin with the magnetic field induced by charges (e.g. nuclei and other electrons) in relative motion.
  - Based on the **magnetic induction**
- With spin-orbit interaction only the total angular momentum is conserved: $\boldsymbol{j} = \boldsymbol{l} + \boldsymbol{s}$.



Uranyl: gross populations from projection analysis in localized orbitals with spin-orbit coupling

ω	Energy	$6s_{1/2}$	$6p_{1/2}$	$6p_{3/2}$	$5f_{5/2}$	$5f_{7/2}$	$6d_{3/2}$	$6d_{5/2}$	$7s_{1/2}$	O_1	$2s_{1/2}$	$2p_{1/2}$	$2p_{3/2}$	O_2	$2s_{1/2}$	$2p_{1/2}$	$2p_{3/2}$
1/2	-3.422	1.02	0.203	0.587	0.106	0.145											
1/2	-3.422	1.02	0.203	0.587	0.106	0.145											
1/2	-3.156		1.51	0.480													
3/2	-2.634			1.983													
1/2	-2.577									1.920							
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3/2	-2.027		0.02	0.06	0.348		0.152									1.31	
3/2	-2.027		0.02	0.06	0.348		0.152						1.31				
1/2	-1.822			0.02		0.323		0.133				1.39					
1/2	-1.822			0.02		0.323		0.133								1.39	
3/2	-1.731				0.131	0.226		0.143									1.41
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Sum		2.04	1.92	3.64	1.17	1.39	0.30	0.552	0	1.920	1.39	2.72		1.92	1.39	2.72	

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- Turning on spin-orbit coupling **further decreases** the importance of U($6p$) in the U≡O chemical bond.

Decomposition of EFG of ^{235}U (I=7/2) in uranyl

$\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}}\rangle^{(A)}_{\mathrm{intra}}$	U	Principal	2.094
		Hybridization	-2.673
	O_1	Principal	-0.457
		Hybridization	0.021
	O_2	Principal	-0.457
		Hybridization	0.021
$\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}}\rangle^{(\mathrm{U},O_1)}_{\mathrm{inter}}$			0.296
$\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}}\rangle^{(\mathrm{U},O_2)}_{\mathrm{inter}}$			0.296
$\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}}\rangle^{(O_1,O_2)}_{\mathrm{inter}}$			-0.002
$\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}}\rangle_{\mathrm{pol}}$			-6.400
$\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}}\rangle$			-7.262
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s-shells	$\simeq 0$
$2p$	-1.101×10^{-3}
$3p$	-2.410×10^{-3}
$3d$	4.201×10^{-3}
$4p$	2.910×10^{-1}
$4d$	3.550×10^{-3}
$4f$	4.980×10^{-5}
$5p$	-2.84
Total core	-3.56
$6p$	8.620
Total semi-core	8.620
$5f$	-2.450
$6d$	-5.080×10^{-1}
Total valence	-2.96
Total	**2.094**

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Beyond the Dailey-Townes model: not only the p orbitals

1. **Inclusion of core orbitals.**

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Beyond the Dailey-Townes model: not only the p orbitals

1. Inclusion of core orbitals.
2. **Inclusion of d and f shells.**

Decomposition of EFG of ^{235}U (I=7/2) in uranyl

$\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}}\rangle^{(A)}_{\mathrm{intra}}$	U	Principal	-3.501
		Hybridization	-2.365
	O_1	Principal	-0.438
		Hybridization	0.018
	O_2	Principal	-0.438
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$\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}}\rangle^{(\mathrm{U},O_1)}_{\mathrm{inter}}$			0.299
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$\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}}\rangle_{\mathrm{pol}}$			-6.810
$\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}}\rangle$			8.227
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$5f_{7/2}$	-2.650
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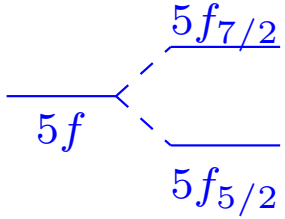
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2. The U($5f$) shell contributes more.

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Huge polarization

Decomposition of EFG of ^{235}U (I=7/2) in uranyl

Chemical information from polarization?

	$\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}} \rangle_{ii}$	Polarization
Core	-3.29	-3.11 (**94.55%**)
Non-bonding U($6p$)/O($2p$)	28.52	4.38 (15.37%)
U≡O_1	-16.25	-3.84 (23.63%)
U≡O_2	-16.25	-3.84 (23.63%)
Electronic: $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}} \rangle$	-7.26	-6.40 (88.14%)
Nuclear: $e\hat{q}^{\mathrm{nucl}}_{\mathrm{U}}$	0.96	
Total $\langle e\hat{q}^{\mathrm{el}}_{\mathrm{U}} \rangle$	-6.30	

- Almost all of **the core is polarized**.

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- Almost all of **the core is polarized**.
- **Chemical information:** how much the AOs have deformed once inserted into the molecule.

Conclusions

1. The electric field gradient at the nucleus is a core property:
   - Combined with experiment, gives **nuclear information**.
   - Sensitive to the chemical environment.
   - Sensitive to relativistic effects.

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1. The electric field gradient at the nucleus is a core property:
   - Combined with experiment, gives **nuclear information**.
   - Sensitive to the chemical environment.
   - Sensitive to relativistic effects.
2. Projection analysis provides a decomposition of expectation values:
   - Intra-atomic contributions tend to be large.
   - Intra-atomic contributions from other centers are to a large extent cancelled by corresponding nuclear contributions.
   - The Dailey-Townes model works well in the presence of light atoms.
   - For heavy systems, it is necessary to go **beyond the Dailey-Townes model** and consider d and f orbitals, the core, and relativistic effects.

With thanks to

HAMP-vQED team



Figure 3: From the left: Trond Saue, Johann Pototschnig, Maen Salman, Jan Brandejs, Gabriele Fabbro, Ryan Benazzouk and Dávid Ferenc.

Thank you for your attention